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Ontology-Boosted Deep Learning for Multi-Label Classification of Arabic Abusive Messages on Social Networks

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Abstract

With the interconnection of today's world and the exponential growth of users' generated data on social network, an unprecedented propagation of abusive content has emerged, particularly within the Arabic-speaking community. Deep learning models have shown promise in tackling this issue, yet they demand a substantial amount of data while having a "black-box" nature and a limited interoperability. To address these deficits, we propose to create a representative ontology of abusive messages in Arabic as a way to replicate the domain knowledge of human beings. As part of a broader study, we have already presented a deep learning multi-label model for detecting Arabic abusive messages. However, it is worth noting that this model necessitates a substantial volume of data to demonstrate its efficacy. Our objective here is to enhance our initial contribution by creating a structured and abstract knowledge representation that aims to enrich and specialize our multi-label model while remedying to the aforementioned problems. Exploiting the ontology allows the extraction of additional features that will be fed to the initial model. These features encompass not only effective semantic knowledge but also textual and linguistic forms. Experimental results showed that our model attained a 91% micro-averaged F1-score, marking a 6-point increase compared to the initial deep learning model. Furthermore, it achieved a precision of 91% and a recall of 89%.

Keywords: Ontology ; Knowledge engineering ; Deep Learning ; BERT ; Classifier Chains ; Multi-label Classification ; Natural Language Processing ; Arabic Language ; Abusive Texts ; Social Media

1. Introduction

Social media is increasingly becoming an omnipresent part of our life. With the easy access and the anonymity privileges it grants to users, people are getting more comfortable expressing their feelings, ideas, and opinions, publicly and instantaneously. Unfortunately, this ease of consuming and disseminating information has led to an alarming spread of abusive content. The same potential of social media brings together a serious danger which is providing an interactive space for abusing any targeted individuals or groups.

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Consequently, solid protection has proven to be necessary and continues to be so, to rapidly detect and effectively remedy such phenomena. In that context, deep neural networks architectures have shown promising results in NLP field and many other ones. That was our motivation point to present, in a study with a broader scope [5], a multi-label classification solution based on deep neural networks and ensemble learning. Good results were achieved despite the modest size of the dataset.

Our paper proposes a novel approach that lies in boosting our initial multi-label classification model by injecting features extracted from an ontology that accurately represents abusive messages in Arabic. Considering the substantial amount of training data it requires, we created an ontology that offers a structured and abstract representation of knowledge specifically designed to enhance our multi-label model for the detection of abusive messages across various forms in Arabic social media. We have meticulously crafted this ontology to encompass properties and relations that define the characteristics and links between the various concepts contained within abusive messages. The aim is to use the textual and linguistic forms found in the messages, coupled with the semantic knowledge provided by the ontological representation to augment the performance of our system and boost the deep learning classification as the size of the training data is not large. This mixture of deep learning with semantic knowledge offers a novel approach to understanding and categorizing abusive content. It is particularly valuable given the complexity of the Arabic language and the multitude of abuse manifestations, and it represents an innovative step forward in dealing with multilabel classification issues.
Furthermore, a noteworthy advantage of developing a specialized ontology lies in its inherent reusability and potential for future applicability across a spectrum of related tasks. Despite the initial effort required for its creation, the ontology serves as a foundational resource that can be leveraged repeatedly.

The rest of the paper is organized as follows: Section 2 reviews the related work on combining ontologies with deep learning models in different fields. In section 3, we describe our ontological approach with its key concepts, data, and object properties, as well as the concept dictionaries. Following this, we describe in section 4 the enhancement of our initial deep learning model by elucidating the extraction process of ontological features, along with the feeding process. Then, in section 5, we go through the experimentation and evaluation performed, before concluding with a discussion of the achieved results in the last section (section 6).

2. Related works

The idea of combining ontology reasoning with deep learning models was presented in [12] for remote sensing image semantic segmentation. Motivated by the interpretability and reliability issues inherent to data-driven deep learning methods, the authors proposed an approach where deep learning and ontology reasoning modules mutually enhance each other. Furthermore, the goal pursued by [4] and [11] was to provide interpretations for machine learning outcomes, while [7] aimed to counter the black box effect of deep learning models.

In other contexts, [10] suggested using domain ontologies as an alternative to classifiers for sentiment analysis on tweets, while [3] employed an ontology-based approach as a training-less classifier, with the novelty lying in dynamically generating the ontology. Meanwhile, [14] utilized rule-based methods on an ontology aiming to detect discriminatory language.
Turning to the domain of abusive language and related areas, [16] introduced an Ontology of Italian Dangerous Speech Messages. This initiative arose from the necessity to provide a structured organization to the myriad linguistic resources and datasets in the NLP field, and to ensure interoperability and dialogue between similar resources.

There exists a variety of perceptions and approaches regarding the integration of ontologies with deep learning. One perspective involves the use of deep learning techniques in the creation of the ontology itself [6]. Another viewpoint involves the creation of a reusable ontology from existing heterogeneous datasets, serving as a global reference for community use. Another approach, which is adopted in our research, consists in creating an ontology that reflects human domain knowledge perception. This ontology is then utilized to augment existing deep learning models by integrating semantic knowledge as additional features into the model ([15]).

3. Ontology-based approach

In computer science, ontology is a formal representation of the knowledge by a set of concepts within a domain and the relationships between those concepts [13]. It provides a formalization of knowledge to ensure data interoperability and facilitate information retrieval and analysis. It acts as a mechanism for semantic modeling and the explicit capture of domain-specific knowledge [17].

Developing an ontology specifically for the concept of abusive messages has numerous benefits. To begin with, it enables us to formalize and structure our comprehension of the domain pertaining to abusive messages. This facilitates a clear understanding of various forms of abuse and how they can occur. Furthermore, it enhances the efficiency of AI models. Incorporating the ontology as supplementary features in our model equips it with contextual data, thereby allowing the model to better comprehend and categorize abusive messages. Even more, an ontology can be, as in our case, exploited in a much less conventional way by extracting textual features and linguistic forms existing in the dataset.

The ontology we developed encompasses eight classes of abusive content namely: Racism, Sexism, Xenophobia, Pornography, Violence, Hate, LGBTQIA hate, and Religious hatred, ensuring comprehensive coverage of various forms of abusive messages. We meticulously designed it to incorporate data properties and relationships that accurately define the characteristics and connections between the concepts within abusive messages. In this section, we will go first into the key concepts that form the basis of this ontology. Then, we will present the data properties that give each concept its unique attributes, and the object properties that define the relationships among these concepts. Finally, we will end the section with the dictionaries related to each concept.

3.1. Key concepts

The foundation of our ontology is built upon several key concepts that we have identified. We note that they are not exaustive and can be enriched. However, they certainly are crucial for comprehensively understanding and representing abusive messages. These concepts are:

1. **Abusive Message:** This forms the primary entity of our study. The abusive message encapsulates the various forms of offensive and harmful communication that occur in the digital space.
2. **Subclasses of Abusive Messages:** These include ”Violent Messages,” ”Adult Messages,” ”Hateful Messages,” and ”Discriminative Messages.” Each of these subclasses signifies a distinct category of abusive content, allowing for more precise categorization and analysis.
3. **Discriminative Message Subtypes:** This concept breaks down the category of discriminative messages into more specific forms of discriminatory abuse, such as ”Racist,” ”Sexist,” ”Religious Hatred,” ”Xenophobic,” and ”Anti-LGBTQIA” messages.
4. **Basis of Discrimination:** This concept refers to the underlying reasons or grounds on which discriminatory messages are based, including belongings, preferences, or physical appearances.
5. **Abusive Words:** These are specific keywords or phrases associated with each subclass of abusive messages. They serve as indicators to help identify the specific type of abuse conveyed in a message.
6. **Sender and Target Entities:** These concepts capture the originator of the abusive message (sender) and the recipient or object of the abuse (target), offering a clearer understanding of the dynamics in abusive interactions.
7. **Implicit Expressions:** This concept acknowledges that abusive messages may not always explicitly convey abuse but may often do so implicitly or indirectly. This helps capture subtler forms of abuse.
8. **Special Characters:** This concept covers the use of special characters often used in abusive messages, either to enhance the abusive intent or to evade detection mechanisms.

3.2. Data properties and object properties

In the context of knowledge representation, data properties are aspects, characteristics, attributes, or parameters that entities or individuals can have. We aim in Table 1 to comprehensively define the data properties pertaining to the main concepts in our ontology being considered with the goal of enhancing our understanding and representation of the model.

Table 1. Concepts and Data Properties of Abusive Messages

Concept	Data Property	Description
Abusive message	Content	The actual content of the message
	Length	The length of the message
	URL	A hyperlink in the message, if any
	Mention	Tags in the message, if any
Target	Name	The name of the target
	Type	The type of the target (An individual, a community, a group, etc.)
	Function	A public figure can have a certain function (Actress, president, soccer player, etc.)
Sender	Name	The name of the sender
	Type	The type of the sender
	Function	A public figure for example can have a certain function (Actress, president of France, soccer player, etc.)
Implicit mode	Mode	The implicit conveyance method.
	Pattern	Indicates the pattern followed in this message to express abuse
Word	Part of Speech	The part of speech of this word
	Definition	The definition of this word
	Synonyms	The set of synonyms for this word
	IsExclamationMark	To indicate if the word is an exclamation mark
	IsQuestionMark	To indicate if the word is a question mark
	SpecialCharacters	The set of special characters in this word

Object properties, on the other hand, within the context of ontology and knowledge representation, are the relations between pairs of individuals. They represent how individuals interact together within a specific domain of knowledge. In this context, we are going to present and describe the object properties between the different entities that make up our ontology of abusive messages, and discuss the properties and specificities of each.

- **HasTarget** (between 'Abusive Message' and 'Target'): This property denotes the individual or group that the abusive message is directed at.
- **HasSender** (between 'Abusive Message' and 'Sender'): This property identifies the individual who sent the abusive message.
- **HasWord** (between 'Abusive Message' and 'Word'): This property links the abusive message to the specific words it contains, facilitating detailed content analysis.
- **Conveys** (between 'Abusive Message' and 'Implicit Mode'): This property signifies the implied or concealed meanings or tones within the abusive message.
- **Contains** (between 'Abusive Message' and 'Special Characters'): This property points out the special characters used in the abusive message, an essential component in comprehending the format or emphasis of the message.
- **HasKeywords**: Each entity from 'Violent Message', 'Racist Message', 'Sexist Message', 'Adult Message', 'Hateful Message', 'LGBTQIA Hate Message', 'Xenophobic Message', to 'Religious Hatred Message' is linked by 'HasKeyword' to its respective word entity (e.g., 'Violent Word', 'Racist Word', etc.). This property serves as a critical link, associating distinct types of messages with their characteristic keywords.
- **BasedOn** (between 'Discriminative Message' and 'Base'): This property associates a discriminative message with its underlying basis.

Figure 1 provides a visual representation of a subset of the object properties we have just enumerated.

Fig. 1. A subset of object properties from the Abusive Language Ontology

3.3. Concept dictionaries

In our work, we adopted the key insights mentioned in [2], which imply the following: Firstly, *certain words carry more importance than others*. This suggests that within our system, some words can provide more informative content regarding the classification subject. Secondly, *words themselves are representative of higher-level mental concepts*. Within a given topic, there is a relatively limited number of concepts, while there exists a potentially infinite number of words and ways to express them. By acknowledging these insights, we acknowledge the significance of word importance and the underlying concepts they represent. To illustrate this, and also building upon the work presented in [2], we generated in figure 2 a cloud of words related to the class *racism*. This visualization not only offers a visible representation of one of our concept dictionaries but also illuminates the range and frequency of terms closely tied to the notion of racism.

Fig. 2. A Word Cloud Exploration of the Racism class

4. Enhancing deep learning model

Having presented the comprehensive ontology designed to capture the domain-specific nuances, textual patterns, and linguistic constructs of abusive messages, we now describe how this ontology can be employed for feature extraction. This process is pivotal, as it will transform our abstract knowledge representation into tangible features. After that, we present the feature injection process for enhancing the predictive capabilities of our initial deep learning model.

4.1. Features extraction

4.1.1. Textual forms

Beyond the conventional use of ontologies to extract domain-specific knowledge, our methodology leverages the structured nature of ontologies to uncover and capitalize on textual features inherent in our dataset. By doing so, we aim to enhance the model's capability to understand not just the content, but also the form and structure of the input data. The selection of textual features in our approach is based on specific assumptions regarding online users behavior. Table 2 details these features and the corresponding assumptions that informed their inclusion

Table 2. Textual Features and Underlying Assumptions

Textual Feature		**Assumption**
Length of the text		Angry people write short messages
Punctuation	Count of special characters	Used to avoid explicit abusive words and potential bans
	Count of exclamation marks	Multiple exclamation marks are common in abusive messages
	Count of question marks	Abusers might seldom use question marks

4.1.2. Linguistic forms

When analyzing text, the nuances and patterns of language play a very important role. These are known as linguistic features that go beyond the words, revealing deeper layers of meaning and intent.
One powerful linguistic tool in text analysis is Named Entity Recognition (NER). It identifies and categorizes words in text that correspond to specific entities, such as names of individuals, organizations, or locations. Essentially, NER is about extracting structured information from unstructured text data. In the context of abusive message detection, NER takes on heightened importance. Abusive messages often contain indicators or targets, which are essentially named entities. Recognizing a named entity as a potential target of abuse provides a dual advantage. Firstly, it aids in detecting abusive content by recognizing patterns common to such messages. Secondly, it provides valuable context regarding whom or what the abuse is directed towards.

4.1.3. Semantic knowledge

In the domain of text analysis, word embeddings have long been recognized for their capability to capture word-level semantics. However, they often fall short in understanding the broader context, especially domain-specific nuances. By using ontology-based features, we are addressing this gap by explicitly injecting domain knowledge into the model. Moreover, one major challenge with deep learning models is their ”black-box” nature. By incorporating ontology-based features, we introduce a degree of interpretability, allowing users or researchers to understand why certain predictions are made. This is crucial, especially in sensitive applications like detecting abusive messages.

4.2. Features injection

The model proposed within the scope of this research in [5] is a multi-label classification and ensemble deep learning approach. It begins by partitioning a multilabel dataset into smaller datasets, each containing the same number of labels. This partitioning is achieved using a dedicated algorithm that fully exploits the correlations between the labels,

organizing the smaller datasets accordingly. After that, a deep learning classifier, which is the multilingual BERT model developed by [8], is trained on each set of labels. All the classifiers are interconnected via the feature space following the correlation order. When classifying a new instance, it traverses through the chain of trained classifiers. For each label, the result provided by the best classifier for that label, which was determined during the training phase, is selected. The overall architecture is described in figure 3 (a).
A key step in our chaining approach is passing classification information from one BERT classifier to the next. Given that classification outputs are not text but binary vectors, a straightforward approach might not be effective as BERT is designed to work with text. To solve this, we were inspired by the idea from [9] and we took the BERT [CLS] token embedding and concatenated the binary features at the end as shown in figure 3 (b). This allows us to pass the classification information to the next BERT classifier in a form that can be efficiently processed.

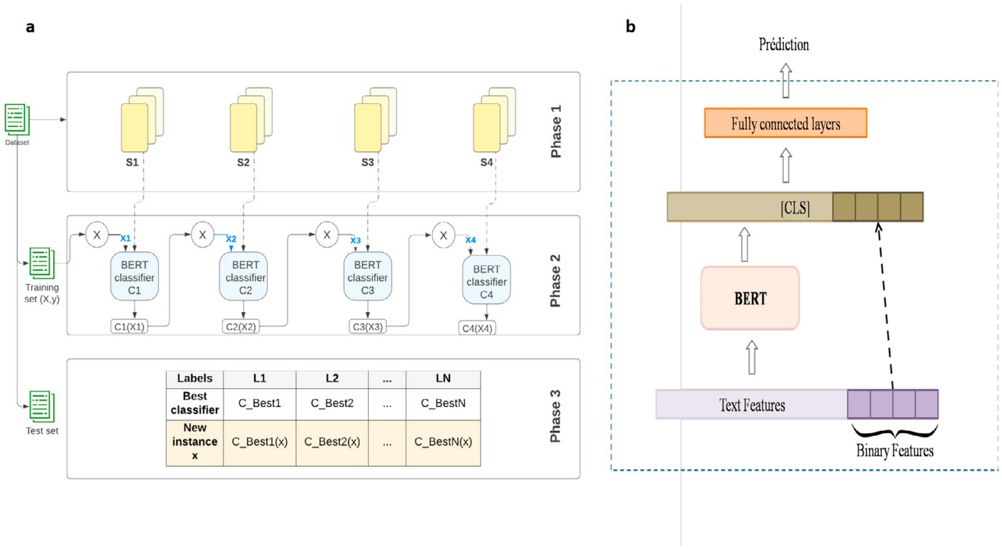

Fig. 3. (a) The architecture of the proposed approach ; (b) Mixing BERT with binary features

The features extracted from our ontology have followed the same concatenation logic depicted in Figure 3, immediately after the features passed in the chain. Each classifier receives semantic features pertinent to the classes it is trained on, along with all textual and linguistic features.

5. Experimentation and evaluation

5.1. Experimentation setup

In this section, we discuss the steps we followed to create our ontology, along with the different tools and setups we implemented.

1. **Implementing the ontology**
   We started our experimentations with the implementation of the ontology using the ”Protégé” tool. We carefully built the various components of the ontology, namely its concepts, data properties, object properties, and concept dictionaries. Once completely developed, we exported the ontology into a ”.ttl” file format.
2. **Python Program and Dataset Integration**
   With the ontology ready, the next step was to develop a Python program capable of reading and understanding it. We loaded our dataset (prior to any preprocessing) into the program to ensure the accurate extraction of textual forms. The program, using the formal representation of knowledge from our ontology, perused the dataset. For each entry, it utilized the ontology to extract varying types of textual features.

3. **Named Entity Recognition using Stanza**
   For our final linguistic feature extraction, we employed Stanza, a Python natural language analysis package. We used it to discern named entities, identifying them as our linguistic feature: the ”Target.”
4. **Integrating sémantic features (Figure 4)**
   For each concept dictionary related to one of the eight classes of an abusive message, we compute the relative embedding vector by averaging the embedding vectors of each word in the list. After that, for each embedding vector (i.e., for each class), we proceed as follows: We iterate over the sentences from the dataset, going word by word. For each word, we compute its embedding and then calculate its cosine similarity with the embedding vector of the class in question. Ultimately, we consider the maximum similarity value as the value for our feature.

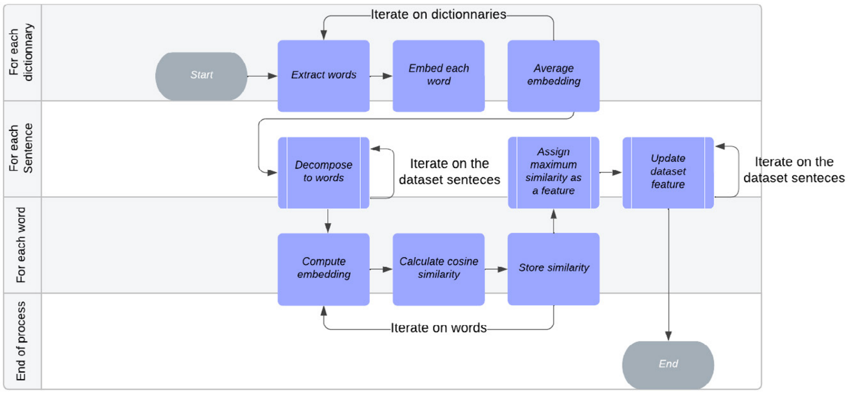

Fig. 4. The workflow of integrating semantic features to the dataset

5. **Normalization**
   Given the diverse nature of our features, it’s evident that they operate on different scales. Some features, for instance, might have values such as 1, 2, or even greater than 10. In contrast, others might fall within a range between 0 and 1. Normalization is the process of scaling individual samples to have a unit norm.
6. **Features significance**
   To determine the significance of various features before adding them to our model, we calculated the correlation measure between the different classes and the extracted textual features and we observed the following: Exclamation Marks are prevalent in different abusive subtypes; specifically, 5 out of the 8 categories showed a high correlation with exclamation marks. Moreover, special characters are predominantly used within contexts related to pornography and LGBTQIA, likely as a means to evade automatic bans. Finally, the number of emojis in messages was surprisingly deemed insignificant for our purposes. Their presence did not correlate with any of the subcategories or the “none” category. Thus, we chose to exclude them from our features.
7. **Feeding the features for the training process**
   For our training process, we incorporated both textual and linguistic features, distributing them equally across all eight classes in our model. Given that our model combines ensemble learning and a chain model, each classifier is trained on a subset of classes. This means that for every classifier, we supply only the semantic features related to the classes it’s training on. However, the common textual and linguistic features are provided universally to all classifiers. The features were concatenated with the [CLS] embedding as in figure 3, following the chain features which are the outputs from the preceding classifier.

5.2. Dataset

The dataset used in our experimentation is the **Multi-Label Multi-Platform Dialectical Arabic Dataset of Abusive Messages** [1], which was constructed as part of a previous research. This dataset contains 6000 lines and eight different classes of abusive messages. It allows for a thorough and specific evaluation of our approach within the context for which it was designed. It is presented in figure 5 where numbers under every label indicate whether the instance belongs to this subclass (Number 1) or not (number 0).

Text	Racism	Sexism	Pornography	Religious hatered	Violence	Lgbtqia hate	Hate	Xenophobia
رب استرنا واستر ولادنا *(Lord, protect us, and protect our children)*	0	0	0	0	0	0	0	0
الثوره الثانيه يجب ان تكن ضد المثلين *(The second revolution should be against homosexuals)*	0	0	0	0	0	1	1	0
قتلوه ورتحنا منو ومن عهر متاعو *(They killed him and relieved us of him and his fornication)*	0	0	0	0	1	0	1	0
شنوه هالكحله مخيب منظرها *(What is this black how ugly she is)*	1	0	0	0	0	0	1	0
الله يلعنك يا ابليس، يعني مليت من روتينك وطلعت لنا جيل المثليين و النسويات. ما يكفي اليهود والكفار والملحدين *(God damn you Satan. You got tired from the routine and brought us a generation of gays and feminists. It wasn't enough with jews, the infidels and the atheists?)*	0	1	0	1	0	1	1	0

Fig. 5. Examples from the multi-label Arabic abusive language dataset

6. Results and Discussion

To evaluate the effectiveness of our system, we compared its performance with our initial contribution. We present results using micro-averaged metrics that aggregate performance across all classes and provide overall measures. Experimental results highlight that our new approach delivers a noteworthy enhancement, recording a 6-point increase in the F1-score. Furthermore, improvements were also observed in the Recall and Precision metrics (Figure 6).

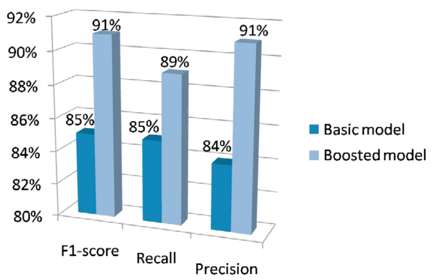

Fig. 6. Overall classification results for the the second contribution

Our findings indicate that while deep learning is undeniably a powerful tool, it demands a vast amount of training data to be effective. Incorporating additional semantic and linguistic knowledge to enhance deep learning proves beneficial based on the approach we have proposed. This concept mirrors real-life learning principles for children. While it's commendable that they learn autonomously, rules and foundational knowledge are equally crucial. However, one must be cautious not to over-rely on these hand-crafted features, lest we revert entirely to traditional machine learning methodologies.

7. Conclusion

In the research presented in this paper, we presented ontology-boosted deep learning approach for the multi-label detection and classification of Arabic abusive on social networks. We meticulously created an ontology that offers a profound and organized representation of abusive content, ensuring thorough coverage of the eight classes which are: Racism, Sexism, Pornography, LGBTQIA hate, Violence, hate, Xenophobia, and Religious hatred. By exploiting the knowledge encapsulated within the ontology, we augmented the deep learning model's feature set, aiming to boost its accuracy in categorizing abusive content. We presented all the steps of our approach, from the ontology's construction and its operational details to its integration into our model. The empirical results confirms the efficiency of integrating structured knowledge in enhancing the model performance.
While our research achieved significant results, it's crucial to acknowledge inherent limitations. The dataset utilized is of modest size and this small scale not only poses challenges for model generalization but can also make the model susceptible to overfitting. Additionally, The implicit nature of some messages is also overlooked. It is a significant aspect where irony, metaphors, and context play key roles. The research presented can be expanded in several directions. For instance, enhancing our model's detection capabilities by adding a high number of real-world examples could lead to a more nuanced understanding.

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